Hypothesis Testing (Pt. II)

Code

STAT 4714

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0.1 Goals for Today

- **Review** the hypothesis testing framework
- Build intuition for **Type I/II errors** and **power**
- Use **simulation-based** tests for **two-sample means**
- Understand the relationship between power and n

0.2 Review: Hypothesis Testing

1. **State** H_0 and H_a in context
2. **Choose** a test statistic (e.g., mean, difference, proportion, quantile, etc.)
3. **Build** the **null distribution** of that statistic assuming H_0
4. **Compute** the p**-value**
5. **Decide** using α (reject or fail to reject) and **interpret in context**

0.3 Will we always decide correctly?

- Every test involves uncertainty; no (reasonable) statistical test gives a guarantee.
- Even if we do things perfectly, we can still *reject a true null* or *fail to reject a false null*.
  - It's important to understand these errors to control risks…

1.1 Type I and Type II Errors

- **Type I (false alarm):** Reject H_0 when it's true.
  - Denoted as α
- **Type II (miss):** Fail to reject H_0 when H_a is true.
  - Denoted as β
- **Power**: Probability we detect a real effect when it exists.
  - Denoted as $1 - \beta$

1.2 Error Table

Type I and Type II Error

Null hypothesis is ...	True	False
Rejected	Type I error False positive Probability = α	Correct decision True positive Probability = 1 - β
Not rejected	Correct decision True negative Probability = 1 - α	Type II error False negative Probability = β

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1.3 Type I & II: PCB Manufacturing

A production line fabricates printed circuit boards designed for $5.0V \pm$ tolerance.

Hypotheses:

- H_0**:** mean voltage $\mu = 5.0V$
- H_a**:** mean voltage $\mu \neq 5.0V$

Power:

- High power means correctly detecting faulty production without excessive downtime.

1.4 Type I & II: PCB Manufacturing

A production line fabricates printed circuit boards designed for $5.0V \pm$ tolerance.

Errors:

- **Type I:** Reject H_0 when it's true → shut down good production
  - *Cost:* lost production time, unnecessary redesign, etc.
- **Type II:** Fail to reject H_0 when H_a is true → continue faulty production
  - *Risk:* producing bad products, customer complaints, etc.

1.5 Type I & II: Router Firmware

A new firmware should keep packet loss $\leq 1\%$.

Hypotheses:

- H_0**:** True packet loss rate $= 1\%$
- H_a**:** Packet loss $> 1\%$

Power:

- High power means catching underperforming firmware before release.

1.6 Type I & II: Router Firmware

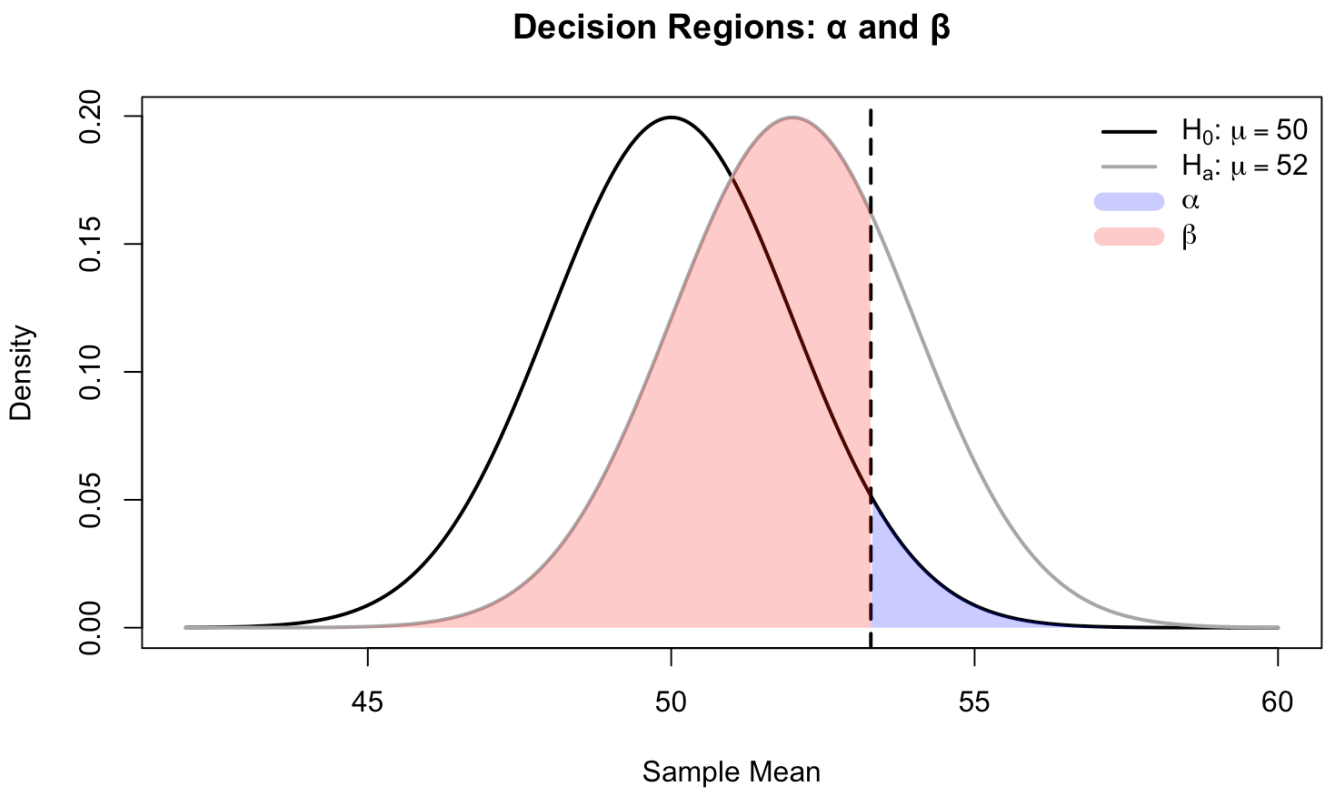
A new firmware should keep packet loss $\leq 1\%$.

Errors:

- **Type I:** Reject H_0 when it's true → scrap a reliable firmware build
  - *Cost:* slower rollout, lost opportunity, etc.
- **Type II:** Fail to reject H_0 when H_a is true → release firmware with real packet loss $> 1\%$
  - *Risk:* degraded network performance, customer complaints, etc.

1.7 Visualizing Decision Errors

H_0: $\mu = 50 \qquad H_a$: $\mu = 52$



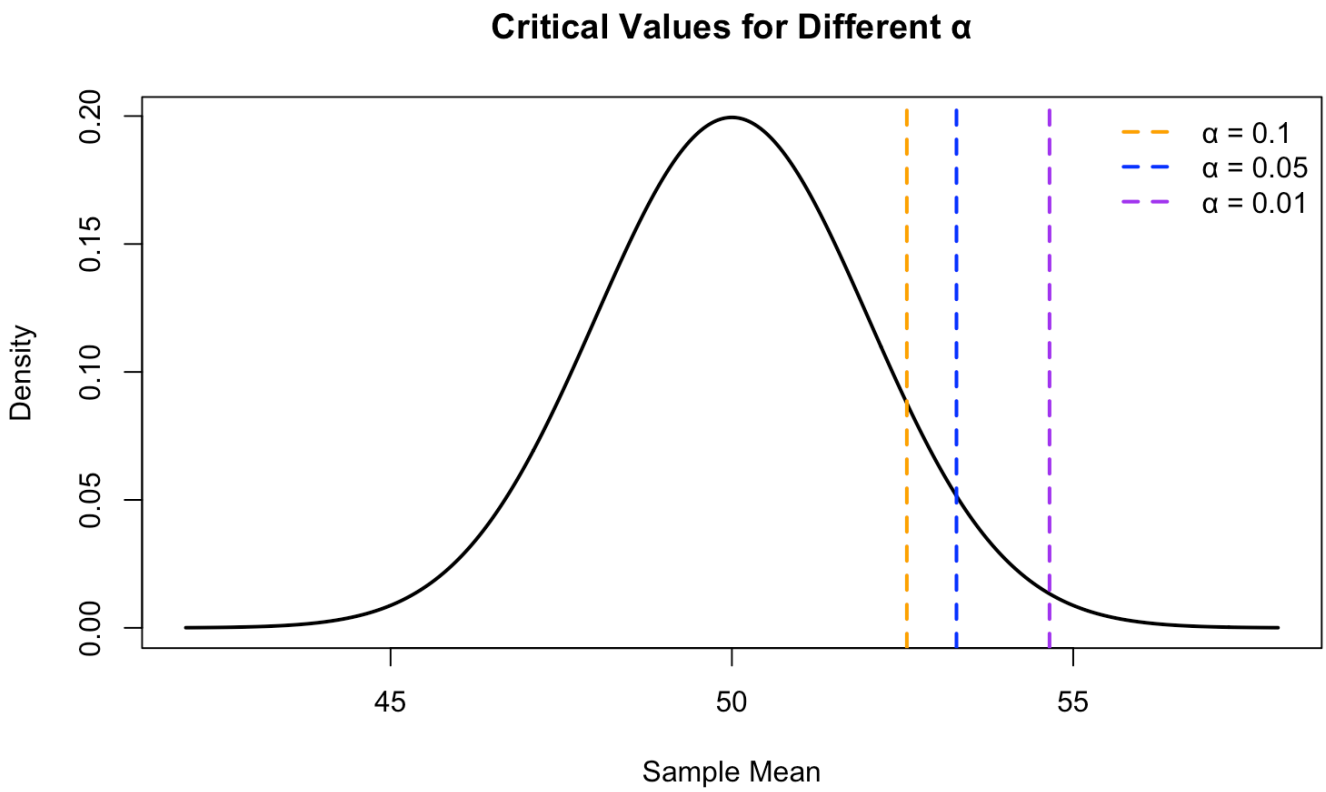
An upper-tailed test. The blue region (α) is the area under the H_0 curve to the right of the critical value (vertical dashed line). The red region (β) is the area under the H_a curve to left of the cutoff. Power is the rest of the area under the H_a curve to the right of the critical value.

1.8 Interpreting the Graphic

- **Blue (α):** Chance of a **false alarm** — rejecting a stable $\mu = 50V$ production line; depends on H_0
- **Red (β):** Chance of a **miss** — keeping the line running even though it's shifted to $\mu = 52V$; depends on H_a
- **Power** $= 1 - \beta$**:** The probability we *catch* the shift correctly; depends on H_a
- **Importantly**, the critical value slices both curves. Because the curves overlap, mistakes are inevitable. By choosing α and planning n, we can control the balance between false alarms and misses…

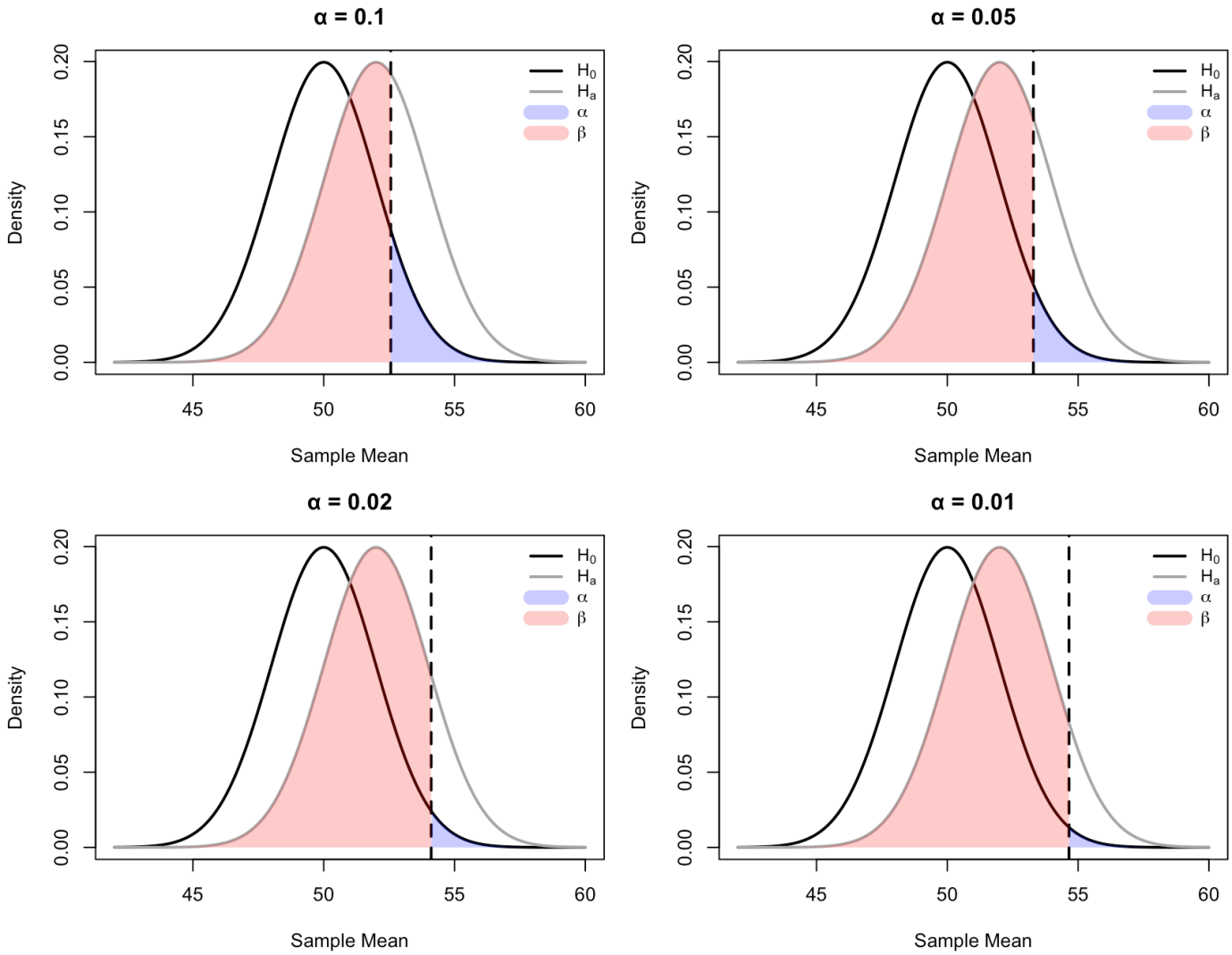
1.9 How α Affects Decision Errors

- Smaller α → harder to reject H_0 → fewer false alarms (↓ Type I) but potentially more misses (↑ Type II)
- Larger α → easier to reject H_0 → more false alarms (↑ Type I) but less misses (↓ Type II).



Lowering α moves the critical value right (harder to reject).

1.10 How α Affects Decision Errors



Upper-tailed mean tests across four α levels. Blue = Type I error (α) under H_0; Red = Type II error (β) under H_a.

1.11 Confidence & Errors

- Recall that our **confidence level** is $1 - \alpha$.
- Probability of rejecting a true H_0 (Type 1 Error rate) $= \alpha$.
- For a fixed sample size n, **lowering** α (increasing confidence) **raises** β (more misses) and **reduces power** $(1 - \beta)$.
- Conversely, **raising** α (decreasing confidence) **lowers** β and **increases power**, but yields more false alarms.
- To improve both (↓α and ↓β), increase n!

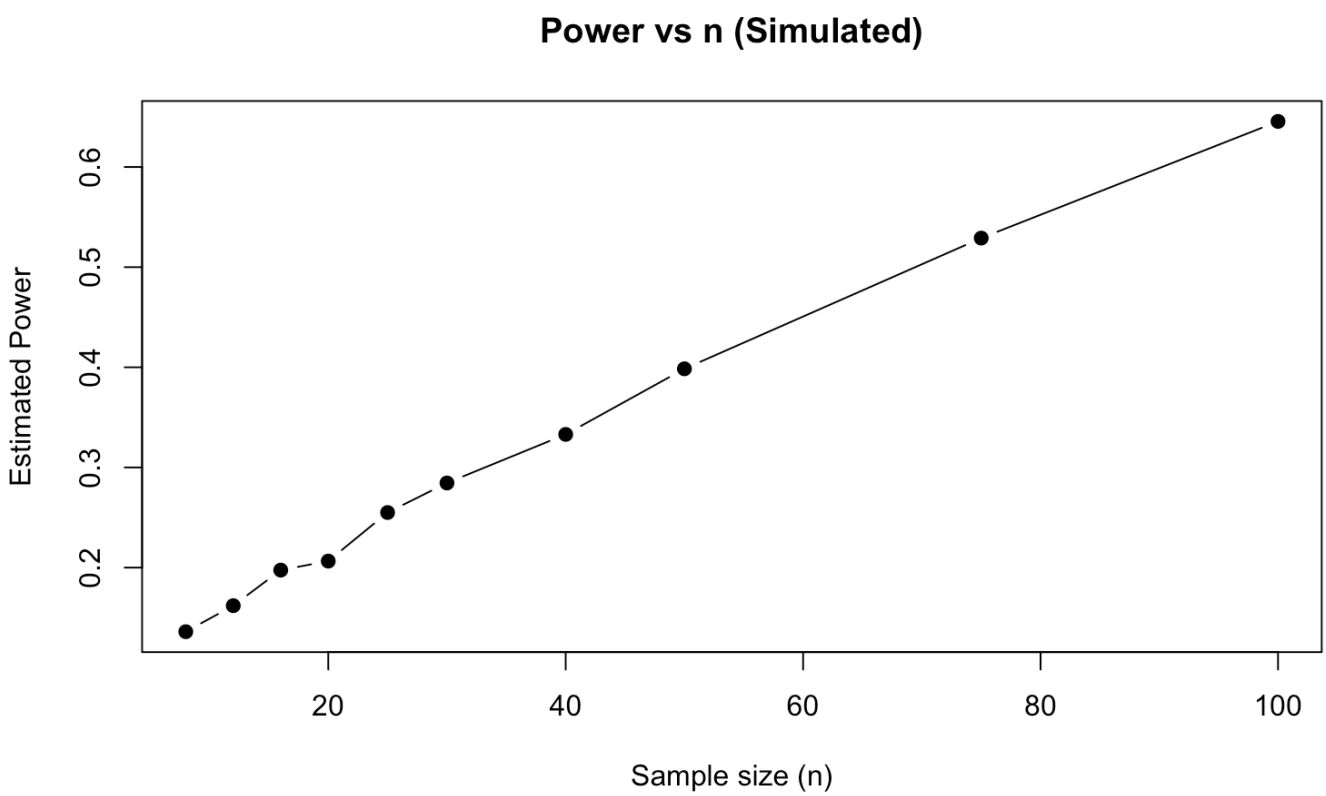
1.12 Why Power Increases with n

- Bigger n → **less noise/more precision** → easier to spot true effects
- As n grows, the alternative distribution **separates** from the null, so β **decreases** and **power** $= 1 - \beta$ **increases**.

- Let's see an example…

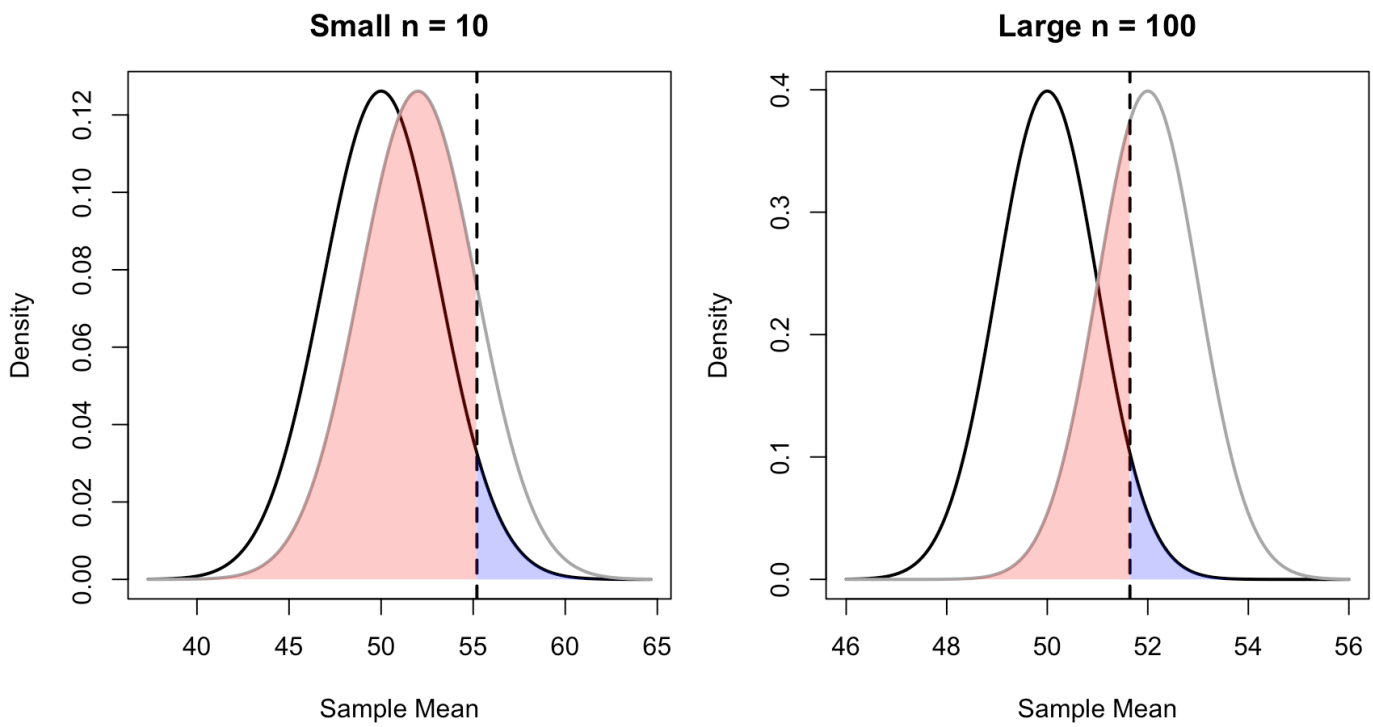
1.13 Sim: Why Power Increases with n

- Fix $\mu_0 = 50, \mu_A = 52, \sigma = 10, \alpha = 0.05$.
- For each n, compute the α region under H_0. Then, **simulate under** H_a and record the rejection rate (power).



Simulated power increases with sample size (for a right-tailed mean test).

1.14 Sim: Why Power Increases with n



Smaller n (left) has wider curves and, thus, more overlap (higher β). Larger n (right) has tighter curves and, thus, less overlap (lower β). Blue = α under H_0; Red = β under H_a.

2.1 Ex: Mean Current Draws

You measure idle current (mA) on **Design A** vs **Design B** and care about the difference in means $\Delta = \mu_A - \mu_B$. Suppose we know from prior testing that $\sigma_A = \sigma_B = 0.7$ and the currents are normally distribution. With samples of size $n_A = 32$ and $n_B = 41$, we see that $\bar{X}_A = 3.2$ and $\bar{X}_B = 2.9$.

Hypotheses:

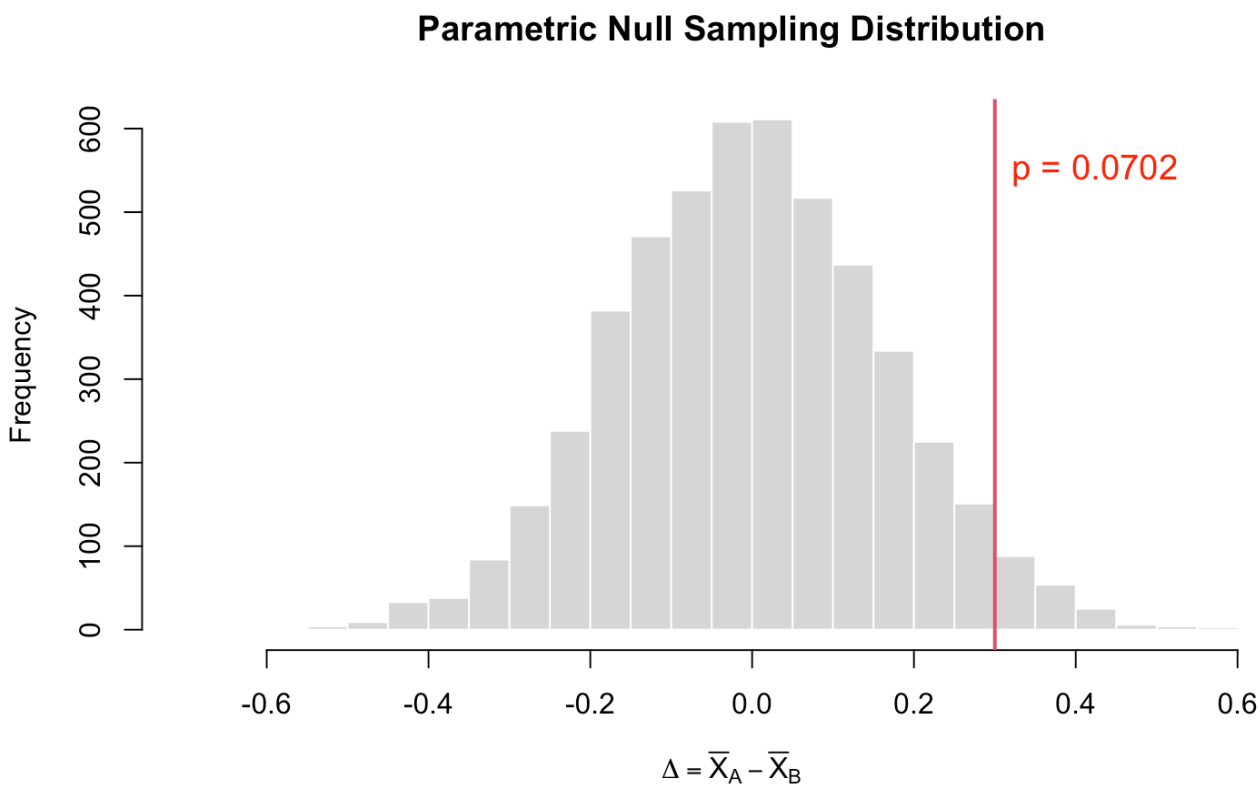
- $H_0 : \Delta = 0$ (no difference)
- $H_a : \Delta \neq 0$

2.2 Ex: Mean Current Draws

- If we **reject** H_0, there is evidence that mean current draw differs (i.e., there is a *better* design).
- If we **fail to reject**, the data are consistent with no difference; consider **increasing** n to improve power if detecting small differences is important.
- **Type I error**: Switching designs when there's truly **no** benefit (e.g., manufacturing/engineering cost).
- **Type II error**: Keeping inferior design when a better one exists (e.g., battery life impact).

2.3 Ex: Mean Current Draws

▶ Code



Parametric null sampling distribution for Δ with group-specific SDs. Red = observed Δ.

2.4 When Parametric Simulation Works…

- In this example, we assumed normal distributions with known σ, which let's us simulate directly from the null models.
  - We call this the **parametric** route!
- But in practice, we may not know the right distribution – or σ may not be known across groups.
- Additionally, computing α and power for two-sample tests is difficult using this parametric route.

2.5 … And When It Doesn't

- We can still test hypotheses by simulating under H_0, but without assuming a specific model!
- The trick: we use the **observed data itself** as the basis for resampling – not some hypothesized distribution (i.e., H_0)
  - *Permutation tests*: shuffle group labels to erase differences.
  - *Bootstrap tests*: recenter or resample so H_0 holds.
  - These approaches let us build a null distribution even when parametric assumptions break.

3.1 Summary

- Same hypothesis testing **framework** across problems
- **Decision errors** are unavoidable; manage α and design for **power**.
- **Simulation under** H_0 provides a flexible, intuitive path for two-sample tests – no extra formulas are required.
  - However, our parametric sampling approach requires knowledge about the hypothesized distributions **and** computing α and power for two-sample tests is tricky.
    - Next, we will learn easier ways to work with two-sample tests!

3.2 Breakout Work

1. For the current draw example, write H_0, H_a, the test statistic, and which tail(s) you'd use.
2. For the router firmware example, define Type I and Type II errors and the potential costs in your own words.
3. Explain the logic in the two-sample simulation code. Report the p-value and interpret in your own words.
4. If you doubled sample size, explain how and *why* power changes without doing any math.